

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286974

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Matsumoto; Akira

SHEET DISCHARGING APPARATUS, IMAGE READING APPARATUS, AND IMAGE FORMING APPARATUS

Abstract

An image reading apparatus includes a support tray on which a sheet is supported, a feeding portion to feed the sheet supported on the support tray, a reading portion to read an image from the sheet fed by the feeding portion, and a discharging portion to discharge the sheet, from which the image has been read by the reading portion, in a sheet discharge direction. In addition, a stack tray on which the sheet discharged by the discharging portion is stacked is provided, and a swing member is supported swingably by the support tray and comes in contact with an upper surface of the sheet stacked on the stack tray. The stack tray includes a first rib extending in the sheet discharge direction and comes in contact with the swing member.

Inventors: Matsumoto; Akira (Toride-shi, JP)

Applicant: CANON KABUSHIKI KAISHA (Tokyo, JP)

Family ID: 73550929

Appl. No.: 19/217295

Filed: May 23, 2025

Foreign Application Priority Data

JP 2019-102798 May. 31, 2019

Related U.S. Application Data

parent US continuation 18582956 20240221 parent-grant-document US 12335447 child US 19217295

parent US continuation 18104808 20230202 parent-grant-document US 11943412 child US 18582956

parent US continuation 17825039 20220526 parent-grant-document US 11595541 child US 18104808

Publication Classification

Int. Cl.: **H04N1/12** (20060101); **B65H31/00** (20060101); **G03G15/00** (20060101); **H04N1/00** (20060101)

U.S. Cl.:

CPC **H04N1/121** (20130101); **B65H31/00** (20130101); **G03G15/607** (20130101);
H04N1/00554 (20130101); **B65H2404/63** (20130101)

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a sheet discharging apparatus that discharges a sheet, an image reading apparatus that reads image information from a sheet, and an image forming apparatus that forms an image on a sheet.

Description of the Related Art

[0002] An image reading apparatus incorporated in an image forming apparatus such as a copier or a multifunctional apparatus includes an auto document feeder: ADF that automatically feeds one sheet at a time serving as a document. A sheet fed by the ADF is supported on a discharge tray after image information thereof is read by an image sensor.

[0003] In recent years, there has been a demand for automatically feeding sheets of various sizes by an ADF and reading image information. According to this, there has been a demand that sheets stacked on a discharge tray are precisely aligned also in the case of feeding sheets of different kinds having different lengths in a conveyance direction by the ADF. Japanese Patent Laid-Open No. 2018-64156 discloses a configuration in which an inclined surface inclined such that a more downstream portion thereof in a sheet discharge direction extends more upward is provided on a discharge tray.

[0004] However, it is sometimes difficult to maintain good alignment of sheets of a plurality of sizes even in the case where the inclined surface described in the document above is provided.

SUMMARY OF THE INVENTION

[0005] The present invention provides a sheet discharging apparatus, an image reading apparatus and an image forming apparatus that can achieve good alignment of sheets of various sizes.

[0006] According to one aspect of the invention, a sheet discharging apparatus includes: a discharge unit configured to discharge a sheet in a discharge direction; a stacking portion on which sheets discharged by the discharge unit are stacked; and a swing guide configured to swing about a swing shaft positioned above the stacking portion. The stacking portion includes: an upstream supporting portion configured to support a sheet discharged from the discharge unit; a wall surface portion erecting upward from an upstream end of the upstream supporting portion in the discharge direction; a first supporting portion provided downstream of the upstream supporting portion in the discharge direction and including a first inclined surface inclined upward toward a downstream side in the discharge direction; and a second supporting portion provided downstream of the upstream supporting portion in the discharge direction and including a second inclined surface inclined upward toward the downstream side in the discharge direction. The swing guide is

configured to overlap with the first supporting portion as viewed in a width direction of a sheet perpendicular to the discharge direction in a case where the swing guide is not in contact with a sheet discharged by the discharge unit, and swing upward in a case where the swing guide is pushed by a sheet. As viewed in the width direction, the stacking portion is configured to satisfy $L1 < L2$, $L4 < L2$, $H1 < H2$, and $H4 < H2$, where $L1$ represents a distance from a first intersection point to a first vertex in a horizontal direction, the first intersection point being an intersection point where the wall surface portion and the upstream supporting portion intersect, the first vertex being a highest point of the first inclined surface in a gravity direction, $H1$ represents a distance from the first intersection point to the first vertex in the gravity direction, $L2$ represents a distance from the first intersection point to a second vertex in a horizontal direction, the second vertex being a highest point of the second inclined surface in the gravity direction, $H2$ represents a distance from the first intersection point to the second vertex in the gravity direction, $L4$ represents a distance from the first intersection point to a second intersection point in the horizontal direction, the second intersection point being an intersection point where the swing guide and the first inclined surface of the first supporting portion intersect, and $H4$ represents a distance from the first intersection point to the second intersection point in the gravity direction.

[0007] According to another aspect of the invention, a sheet discharging apparatus includes: a discharge unit configured to discharge a sheet in a discharge direction; a swing guide configured to swing about a swing shaft; an upstream supporting portion configured to support a sheet discharged from the discharge unit; a wall surface portion erecting upward from an upstream end of the upstream supporting portion in the discharge direction; a first supporting portion provided downstream of the upstream supporting portion in the discharge direction and including a first inclined surface inclined upward toward a downstream side in the discharge direction; and a second supporting portion provided downstream of the upstream supporting portion in the discharge direction and including a second inclined surface inclined upward toward the downstream side in the discharge direction. The swing guide overlaps with the first inclined surface of the first supporting portion as viewed in a width direction of a sheet perpendicular to the discharge direction in a case where the swing guide is not in contact with a sheet discharged by the discharge unit. The swing guide swings upward in a case where the swing guide is pushed by a sheet. A highest point of the second inclined surface is downstream of a highest point of the first inclined surface in the discharge direction. The highest point of the second inclined surface is higher than the highest point of the first inclined surface.

[0008] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a perspective view of a discharge tray of an ADF according to an exemplary embodiment.

[0010] FIG. 2 is a diagram schematically illustrating an image forming apparatus according to the exemplary embodiment.

[0011] FIG. 3 is a diagram schematically illustrating an image reading apparatus according to the exemplary embodiment.

[0012] FIG. 4 is a section view of the image reading apparatus according to the exemplary embodiment.

[0013] FIG. 5 is a section view of a discharge tray according to the exemplary embodiment.

[0014] FIG. 6 is a section view of the discharge tray according to the exemplary embodiment.

[0015] FIG. 7 is a section view of the discharge tray according to the exemplary embodiment.

[0016] FIG. **8** is a section view of the discharge tray and a swing guide according to the exemplary embodiment.

[0017] FIG. **9** is a section view of the discharge tray of the exemplary embodiment illustrating a state in which a document of a first size is supported on the discharge tray.

[0018] FIG. **10** is a section view of the discharge tray of the exemplary embodiment illustrating a state in which a document of a second size is supported on the discharge tray.

[0019] FIG. **11** is a section view of the discharge tray of the exemplary embodiment illustrating a state in which a document of a third size is supported on the discharge tray.

DESCRIPTION OF THE EMBODIMENTS

[0020] An exemplary embodiment of the present invention will be described below with reference to drawings.

[0021] FIG. **1** is a perspective view of an ADF **300** according to an exemplary embodiment of the present disclosure. To be noted, in FIG. **1**, illustration of a feed tray is omitted such that a discharge tray **303** is visible. FIG. **2** is a schematic view of an image forming apparatus **200** including an image reading apparatus **201**. FIG. **3** is a schematic view of the ADF **300** and a reader portion **301** constituting the image reading apparatus **201**. FIG. **4** is a diagram illustrating a sectional configuration of the image reading apparatus **201**.

Image Forming Apparatus

[0022] First, a schematic configuration of the image forming apparatus **200** of an electrophotographic system including the image reading apparatus **201** will be described with reference to FIG. **2**. To be noted, the image forming apparatus **200** is merely an example of an image forming apparatus, and examples of image forming apparatus to which the technique of the present disclosure is applicable include facsimile apparatuses and multifunctional apparatuses including the image reading apparatus **201**. In addition, an image forming unit incorporated in the image forming apparatus is not limited to an electrophotographic system, and the image forming apparatus may include, for example, a printing unit of an inkjet system.

[0023] As illustrated in FIG. **2**, the image forming apparatus **200** includes an image forming apparatus body **202** and the image reading apparatus **201** mounted on an upper portion of the image forming apparatus body **202**. The image forming apparatus body **202** includes an image forming portion **3** serving as an image forming unit disposed approximately at the center thereof, and a feeding unit for feeding a recording material **S** including feed cassettes **6** is positioned below the image forming portion **3**. As the recording material **S**, a wide variety of sheets of different sizes and materials such as paper sheets like regular paper sheets and cardboards, plastic films, cloths, surface-treated sheet materials such as coated paper, and sheet materials having irregular shapes such as envelopes and index sheets can be used. The image reading apparatus **201** including image sensors **409** and **410** serving as image reading units for reading an image of a document is provided above the image forming apparatus body **202**.

[0024] In the image forming apparatus body **202**, the image forming portion **3** is configured as a print engine of an electrophotographic system. The image forming portion **3** of the present exemplary embodiment is configured as a tandem-type intermediate transfer system, and includes four image forming units **10Y**, **10M**, **10C**, and **10K** and an intermediate transfer belt **23** serving as an intermediate transfer member.

[0025] The image forming unit **10Y** forms a yellow toner image by an electrophotographic process. That is, a photosensitive drum **11** serving as a photosensitive member rotates, and a charging device **12** uniformly charges the surface of the photosensitive drum **11**. A laser scanner **13** irradiates the photosensitive drum **11** with laser light modulated on the basis of image information, and thus draws an electrostatic latent image on the surface of the photosensitive drum **11**. A developing unit **14** supplies charged toner particles to the photosensitive drum **11**, and thus develops the electrostatic latent image on the surface of the photosensitive drum **11** into a toner image. This toner image is transferred onto the intermediate transfer belt **23** through primary transfer by a

primary transfer roller **15**. Attached matter such as transfer residual toner remaining on the photosensitive drum **11** without being transferred onto the intermediate transfer belt **23** is removed by a drum cleaner **16**. The process described above is performed in parallel by each of the image forming units **10Y** to **10K**, and toner images of respective colors of yellow, magenta, cyan, and black are formed.

[0026] The intermediate transfer belt **23** is rolled around a plurality of rollers including a secondary transfer inner roller **18**, and is rotationally driven in a direction following a rotation direction of photosensitive drums **11**, that is, a clockwise direction in FIG. **2**. Toner images of respective colors formed by the image forming units **10Y** to **10K** are transferred through primary transfer so as to be superimposed on one another, and thus a full-color toner image is formed on the intermediate transfer belt **23**. This toner image is conveyed to a secondary transfer portion formed between the secondary transfer inner roller **18** and a secondary transfer roller **19** opposed thereto by the rotation of the intermediate transfer belt **23**.

[0027] The image forming apparatus **200** includes a cassette feeding portion **4** and a manual feeding portion **5** as sheet feeding apparatuses for feeding the recording material S. The cassette feeding portion **4** includes the plurality of feed cassettes **6**, and the feeding unit **7** feeds the recording material S one by one from one of the feed cassettes **6** toward a registration roller **17**. In addition, the manual feeding portion **5** provided on a side portion of the apparatus body **202** feeds the recording material S one by one by a feeding unit **8** toward the registration roller **17**. The feeding units **7** and **8** each include a feeding member such as a feed roller that delivers out the recording material S from a feed cassette **6** or a manual feed tray, and a separation member such as a separation roller or a separation pad that applies a frictional force to a sheet of the recording material S superimposed on another sheet of the recording material S conveyed by the feeding member to prevent multiple sheets of the recording material S from being conveyed simultaneously.

[0028] The registration roller **17** delivers the recording material S into the secondary transfer portion in synchronization with the formation of the toner image by the image forming portion **3**. The recording material S on which the toner image has been transferred from the intermediate transfer belt **23** through secondary transfer in the secondary transfer portion is conveyed to a fixing unit **21**. The fixing unit **21** applies heat and pressure to the toner image on the recording material S while nipping and conveying the recording material S, and thus fixes the toner image to the recording material S. In the case of duplex printing, the recording material S having passed through the fixing unit **21** is guided to a reverse conveyance path **26**, switched back, and conveyed to the image forming portion **3** in a state in which a first surface and a second surface thereof are reversed, and an image is formed on the second surface of the recording material S. In the case of simplex printing and in the case where image formation on the second surface in the duplex printing is finished, the recording material S having passed through the fixing unit **21** is discharged from the apparatus body **202** by a discharge roller **25**.

[0029] To be noted, FIG. **2** illustrates a configuration in which the recording material S on which an image has been formed is discharged onto a discharge tray disposed on a side surface of the image forming apparatus body **202** or onto a sheet processing apparatus connected to the image forming apparatus body **202**. Instead of this, a so-called in-body discharge configuration in which a space is provided between the image reading apparatus **201** and the image forming apparatus body **202** in the gravity direction to form a body discharge portion onto which a recording material having undergone image formation in the image forming apparatus body **202** is discharged may be employed.

Image Reading Apparatus

[0030] Next, a schematic configuration of the ADF **300** and the reader portion **301** constituting the image reading apparatus **201** will be described with reference to FIG. **3**.

[0031] As illustrated in FIG. **3**, the image reading apparatus **201** includes the ADF **300** that feeds a

plurality of documents one by one by separating one document from another for reading an image on the document by an image sensor while conveying the document. As the document, a wide variety of sheets of different sizes and materials such as paper sheets like regular paper sheets and cardboards, plastic films, cloths, sheet materials whose surface is treated such as coated paper, and sheet materials having irregular shapes such as envelopes and index sheets can be used. In particular, in the present exemplary embodiment, sheets of a small size that is conventionally hardly handled by an ADF, for example, business cards, can be conveyed by the ADF **300**.

[0032] The ADF **300** includes a feed tray **302** on which documents are placed, and a discharge tray **303** serving as a stacking portion onto which documents whose images have been read are discharged and on which the discharged documents are stacked. The reader portion **301** for reading an image of a document conveyed by the ADF **300** or an image of a still document, for example, a thick document such as a book, is provided below the ADF **300**.

[0033] In the description below, as illustrated in FIG. 3, the left-right direction of the image reading apparatus **201** as viewed from a user, that is, as viewed from the front side of the image forming apparatus **200**, will be referred to as an X direction. The front-rear direction of the image reading apparatus **201** perpendicular to the X direction, that is, the main scanning direction or width direction of documents, will be referred to as a Y direction. In addition, the up-down direction of the image reading apparatus **201** perpendicular to both of the X direction and the Y direction, that is, the gravity direction in normal use, will be referred to as a Z direction. The ADF **300** feeds a document placed on the feed tray **302** toward one side in the X direction, and discharges the document toward the other side in the X direction, that is, in a sheet discharge direction of the present exemplary embodiment, onto the discharge tray **303**.

[0034] Next, the inner structure of the image reading apparatus **201** will be described with reference to FIG. 4, which is a section view of the image reading apparatus **201** as viewed in the Y direction. The ADF **300** includes a pickup roller **401**, a separation roller pair **402**, and a plurality of roller pairs **403** to **406** as a plurality of conveyance units that convey a sheet. The pickup roller **401** abuts the uppermost document among documents **400** placed on the feed tray **302**, and sends out the uppermost document toward the separation roller pair **402**. The separation roller pair **402** separates one document **400** from a plurality of documents **400** when a plurality of documents **400** are received from the pickup roller **401**, and conveys the separated document **400**.

[0035] The plurality of conveyance roller pairs convey the document **400** through a reading position while passing the document **400** onto one another. In this case, the reading position is a position where the image sensors **409** and **410** scan the document **400**. Among these conveyance roller pairs, a pulling roller pair **403** conveys the document **400** while pulling out the document **400** from the separation roller pair **402**. A first reading roller pair **404** and a second reading roller pair **405** convey the document **400** such that the document **400** passes through the reading position while stabilizing the position of the document **400** at the reading position so as to improve the reading precision. A discharge roller pair **406** receives the document **400** having passed through the reading position and discharges the document **400** onto the discharge tray **303**.

[0036] A swing guide **411** is swingably supported on a lower portion of the feed tray **302** disposed above the discharge tray **303**. The swing guide **411** has a function of applying a resistance force to the document **400** discharged by the discharge roller pair **406** to restrict excessive movement of the document **400** in the discharge direction, which is rightward in FIG. 4. To be noted, the swing guide **411** may be urged downward by the weight thereof, or may be, for example, urged downward by a spring member attached to a swing shaft **801**.

[0037] The image sensor **409** serving as a first reading unit is provided in the reader portion **301**. When reading an image from a first surface of the document **400** conveyed by the ADF **300**, the image sensor **409** stops at a position where the image sensor **409** opposes a document feeding-reading glass **407**, that is, at a position illustrated in FIG. 4. When reading an image from a still document placed on a platen glass **408**, the image sensor **409** reads the image while moving on a

rail provided in the reader portion **301** in the X direction, which is a sub-scanning direction.
[0038] Further, an image sensor **410** serving as a second reading unit is provided in the ADF **300** at a position opposing the image sensor **409** of the reader portion **301**. Therefore, images on both surfaces of the document **400** conveyed by the ADF **300** can be simultaneously read by using the two image sensors **409** and **410**. To be noted, as the image sensors **409** and **410**, a contact image sensor: CIS, which is a module of an image sensor such as a complementary metal oxide semiconductor: CMOS and an equal-magnification optical system, and a CCD-type image sensor in which a charge-coupled device: CCD and a reduced optical system are combined can be both used.

[0039] As described above, the ADF **300** serving as a sheet discharging apparatus of the present exemplary embodiment has a function of discharging a document whose image has been read onto the discharge tray **303** by the discharge roller pair **406** serving as a discharge unit. Discharge Tray
[0040] Next, a schematic configuration of the discharge tray **303** will be described with reference to FIGS. **1** to **4**. The discharge tray **303** includes a wall surface portion **100**, an upstream supporting surface **101**, a first projection **102**, second projections **103**, and third projections **104**.

[0041] The wall surface portion **100** is erected, i.e., extends upward in the Z direction, from an upstream end of the discharge tray **303** in the discharge direction. The upstream supporting surface **101** serving as an upstream supporting portion of the present exemplary embodiment extends downstream with respect to the wall surface portion **100** in the discharge direction of documents by the discharge roller pair **406**, that is, in the X direction.

[0042] The first projection **102** serving as a first supporting portion of the present exemplary embodiment is provided downstream of the upstream supporting surface **101** in the discharge direction, and projects upward in the Z direction with respect to the upstream supporting surface **101**. The single first projection **102** is provided at an approximately center position of the discharge tray **303** in the Y direction.

[0043] The second projections **103** serving as second supporting portions of the present exemplary embodiment are disposed such that the first projection **102** is interposed therebetween in the Y direction, and have shapes projecting upward in the Z direction as viewed in the Y direction. The third projections **104** serving as third supporting portions of the present exemplary embodiment are also disposed such that the first projection **102** is interposed therebetween in the Y direction, and have shapes projecting upward in the Z direction as viewed in the Y direction.

[0044] The second projection **103** and the third projection **104** on one side (front side of the image reading apparatus) in the Y direction are connected in the X direction to make up a rib-like projection on the upper surface of the discharge tray **303**. Similarly, the second projection **103** and the third projection **104** on the other side (rear side of the image reading apparatus) in the Y direction are also connected in the X direction to make up a rib-like projection on the upper surface of the discharge tray **303**.

[0045] The arrangement of the first supporting portion, the second supporting portions, and the third supporting portions in the Y direction can be appropriately modified. For example, a plurality of first projections **102** may be provided at a plurality of positions in the Y direction. In addition, two second projections **103** and/or two third projections **104** may be disposed on each side of the center position of the discharge tray **303** in the Y direction, that is, four second projections **103** and/or four third projections **104** may be provided in total.

[0046] The swing guide **411** is provided at a position corresponding to the first projection **102** in the X direction and the Y direction in a space between the feed tray **302** and the discharge tray **303**. The swing guide **411** is supported by a bearing portion **302a** of the feed tray **302** positioned above the discharge tray **303** so as to be swingable about the swing shaft **801** extending in the Y direction. Distal end portions **412** of the swing guide **411** on the side farther from the swing shaft **801** have a shape branched into two portions in the Y direction, and are disposed such that the first projection **102** is interposed between one branch to a first side in the Y direction and the other branch to a

second side in the Y direction of the branched shape. Therefore, in a state in which a document is not discharged onto the discharge tray **303** and the swing guide **411** dangles due to its own weight, the distal end portions of the swing guide **411** overlap with the first projection **102** as viewed in the Y direction as illustrated in FIGS. **1** and **4**. In addition, the distal end portions **412** of the swing guide **411** are respectively positioned between the first projection **102** and the second projection **103** on the one side (first side) in the Y direction, i.e., a first projected portion of the second supporting portion, and between the first projection **102** and the second projection **103** on the other side (second side) in the Y direction, i.e., a second projected portion of the second supporting portion.

[0047] To be noted, although only one first projection **102** and one swing guide **411** are provided in the present exemplary embodiment, a plurality of first projections **102** and/or a plurality of swing guides **411** may be provided at a plurality of positions in the Y direction. In this case, the number of the first projections **102** and the number of the swing guides **411** do not have to be equal, and for example, the number of the swing guides **411** may be smaller than the number of the first projections **102**.

Shape of Discharge Tray and Position of Swing Guide

[0048] Shapes of projections **102**, **103**, and **104** corresponding to the first supporting portion, second supporting portions, and third supporting portions of the present exemplary embodiment will be described in detail below with reference to FIGS. **5** to **7**.

[0049] First, positions of points and the like for specifying the shape of the first projection **102** as viewed in the Y direction are defined as follows with reference to FIG. **5**. [0050] A point where the wall surface portion **100** and the upstream supporting surface **101** intersect is defined as a first intersection point **500**. [0051] A surface constituting an upstream surface of the first projection **102** in the discharge direction of documents and inclined upward toward the downstream side in the discharge direction is defined as a first inclined surface **501**. [0052] The highest point of the first inclined surface **501** in the Z direction, that is, a downstream end of the first inclined surface **501** in the discharge direction, is defined as a first vertex **502**. [0053] A point where the first inclined surface **501** and the upstream supporting surface **101** intersect is defined as a third intersection point **503**. [0054] A straight line connecting the third intersection point **503** and the first vertex **502** is defined as a first straight line Ln1. [0055] A distance from the first intersection point **500** to the first vertex **502** in the X direction is defined as L1 [mm]. [0056] A distance from the first intersection point **500** to the first vertex **502** in the Z direction is defined as H1 [mm]. [0057] An angle of the first straight line Ln1 with respect to the X direction is defined as $\theta 1$ [deg].

[0058] Here, in the case where the upstream surface and the downstream surface of the first projection **102** do not intersect with each other with an angle therebetween but are smoothly connected to each other via a curved surface, the first intersection point **500**, the third intersection point **503**, and the first vertex **502** are virtual points where extension lines of the upstream surface and the downstream surface intersect. This also applies to points for specifying the shapes of the second projections **103** and the third projections **104** that will be described below.

[0059] In addition, in the present exemplary embodiment, since the first inclined surface **501** is constituted by a single flat surface, the first inclined surface **501** coincides with the first straight line Ln1. However, it should be noted that, in the case where the first inclined surface **501** is a curved surface or made up of a combination of a plurality of flat surfaces, the first inclined surface **501** does not coincide with the first straight line Ln1. This also applies to second inclined surfaces **600** of the second projections **103** and second straight lines Ln2 that will be described later.

[0060] Positions of points and the like for specifying the shape of each of the second projections **103** as viewed in the Y direction are defined as follows with reference to FIG. **6**. [0061] A surface constituting an upstream surface of the second projection **103** in the discharge direction of documents and inclined upward toward the downstream side in the discharge direction is defined as a second inclined surface **600**. [0062] The highest point of the second inclined surface **600**, that is,

a downstream end of the second inclined surface **600** in the discharge direction, is defined as a second vertex **601**. [0063] A point where the second inclined surface **600** and the upstream supporting surface **101** intersect is defined as a fourth intersection point **602**. [0064] A straight line connecting the fourth intersection point **602** and the second vertex **601** is defined as a second straight line Ln2. [0065] A distance from the first intersection point **500** to the second vertex **601** in the X direction is defined as L2 [mm]. [0066] A distance from the first intersection point **500** to the second vertex **601** in the Z direction is defined as H2 [mm]. [0067] An angle of the second straight line Ln2 with respect to the X direction is defined as $\theta 2$ [deg].

[0068] As described above, these intersection points and highest points may be virtual points. In addition, in the present exemplary embodiment, although the second inclined surface **600** coincides with the second straight line Ln2 because the second inclined surface **600** is a single flat surface, the second inclined surface **600** may be a curved surface or a combination of a plurality of flat surfaces, and in this case, the second inclined surface **600** does not coincide with the second straight line Ln2.

[0069] Positions of points and the like for specifying the shape of each of the third projections **104** as viewed in the Y direction are defined as follows with reference to FIG. 7. [0070] A surface constituting an upstream surface of the third projection **104** in the discharge direction of documents and inclined upward toward the downstream side in the discharge direction is defined as a third inclined surface **700**. [0071] The highest point of the third inclined surface **700**, that is, a downstream end of the third inclined surface **700** in the discharge direction, is defined as a third vertex **701**. [0072] A distance from the first intersection point **500** to the third vertex **701** in the X direction is defined as L3 [mm]. [0073] A distance from the first intersection point **500** to the third vertex **701** in the Z direction is defined as H3 [mm].

[0074] As described above, these intersection points and highest points may be virtual points.

[0075] Next, the position of the swing guide **411** will be described with reference to FIG. 8. FIG. 8 illustrates a state in which the swing guide **411** is not pushed by a document discharged onto the discharge tray **303**. That is, the swing guide **411** is positioned at a lower limit position in a movable range thereof in a swing direction **802** about the swing shaft **801**. In this state, the distal end portions of the swing guide **411** overlap with the first projection **102** as viewed in the Y direction. More specifically, an upstream surface of the swing guide **411** in the discharge direction intersects with the first inclined surface **501**, which is an upstream surface of the first projection **102** in the discharge direction.

[0076] Positions of points and the like for specifying the shape of the swing guide **411** as viewed in the Y direction are defined as follows. [0077] A point where the first inclined surface **501** and the swing guide **411** intersect is defined as a second intersection point **800**. [0078] A distance from the first intersection point **500** to the second intersection point **800** in the X direction is defined as L4 [mm]. [0079] A distance from the first intersection point **500** to the second intersection point **800** in the Z direction is defined as H4 [mm].

[0080] Further, the shape of the discharge tray **303** according to the present exemplary embodiment is configured such that the values L1 to L4, H1 to H4, $\theta 1$, and $\theta 2$ related to the first projection **102**, the second projections **103**, the third projections **104**, and the swing guide **411** satisfy the following relationships.

[00001] $L1 < L2 < L3$ (1) $L4 < L2$ (2) $H1 < H2 < H3$ (3) $H4 < H2$ (4)

[0081] (1) indicates that the first vertex **502** of the first projection **102**, the second vertices **601** of the second projections **103**, and the third vertices **701** of the third projections **104** are arranged in this order from the upstream side to the downstream side in the discharge direction of documents.

(3) indicates that the first vertex **502** of the first projection **102**, the second vertices **601** of the second projections **103**, and the third vertices **701** of the third projections **104** are arranged in this order from the lower side to the higher side in the gravity direction.

[0082] (2) indicates that the second intersection point **800** where the swing guide **411** and the first inclined surface **501** intersect is positioned upstream of the second vertices **601** of the second projections **103** in the discharge direction. (4) indicates that the second intersection point **800** is positioned lower than the second vertices **601** of the second projections **103** in the gravity direction. As described above, since $L4 < L2$ and $H4 < H2$ are satisfied, the discharge space above the discharge tray **303** is blocked by the swing guide **411** in a state in which the swing guide **411** is not pushed up by a document as illustrated in FIG. **8**. In other words, the downstream side in the discharge direction of the space defined between the discharge tray **303** and the feed tray **302** in the gravity direction as viewed in the Y direction is closed by the first inclined surface **501** of the first projection **102** and the swing guide **411**.

Length of Document in Conveyance Direction and Dimensions of Discharge Tray

[0083] The ADF **300** of the present exemplary embodiment is configured to be capable of handling at least three document sizes having different lengths in the conveyance direction. In the description below, among representative document sizes that can be handled by the ADF **300**, a size having a small length in the conveyance direction will be referred to as a first size, a size having a medium length in the conveyance direction will be referred to as a second size, and a size having a large length in the conveyance direction will be referred to as a third size. For example, the first size is A6R having a length of 148 mm in the conveyance direction and a width of 105 mm. For example, the second size is A5R having a length of 210 mm in the conveyance direction and a width of 148 mm. For example, the third size is A4R having a length of 297 mm in the conveyance direction and a width of 210 mm. To be noted, A6R, A5R, A4R are merely examples, and may be changed to document sizes assumed to be frequently used.

(A) Arrangement of Document of First Size and First Projection

[0084] The first projection **102** of the discharge tray **303** is provided at a position corresponding to a document of the first size. That is, the first projection **102** is disposed so as to come into contact with a leading end of a document in the discharge direction on the first inclined surface **501** in a state in which a trailing end of the document has been released from the discharge roller pair **406** in the case where a document of the first size is discharged. In the case where A6R is assumed as the first size, the first vertex **502** of the first projection **102** is preferably disposed such that $L1$ [mm] and $H1$ [mm] satisfy $148 \leq L1 < 180$ and $20 \leq H1 < 40$.

[0085] In addition, the swing guide **411** is disposed such that a space in which documents of the first size can be stacked is secured on the upstream side in the discharge direction in a state in which the swing guide **411** is not pushed up and overlaps with the first projection **102** as viewed in the Y direction. For example, the swing guide **411** is preferably disposed such that $L4$ [mm] and $H4$ [mm] defining the position of the second intersection point **800** satisfy $L1 = L4$ and $H1 = H4$. To be noted, in the case where A6R is assumed as the first size, even if $L4$ or $H4$ is not equal to $L1$ or $H1$, the swing guide **411** is preferably disposed such that $148 \leq L4 < 180$ and $20 \leq H4 < 40$ are satisfied.

[0086] In the case where the first projection **102** and the swing guide **411** are disposed as described above, the position of a first-size document **900** discharged onto the discharge tray **303** is regulated by the first inclined surface **501** of the first projection **102** and the swing guide **411** as illustrated in FIG. **9**. That is, the first-size document **900** comes into contact with the first inclined surface **501** or the swing guide **411** on a leading end thereof in a state in which a trailing end thereof in the discharge tray has been released from the discharge roller pair **406**. At this time, the swing guide **411** resists an inertia force **901** of the document **900** and basically does not swing to a position above the first projection **102** even when the swing guide **411** comes into contact with the document **900** not nipped by the discharge roller pair **406**. Therefore, downstream movement of the first-size document **900** beyond the second intersection point **800** in the discharge direction is hindered by the swing guide **411** and the first projection **102**. As a result, the first-size document **900** is supported on the upstream supporting surface **101** and the first inclined surface **501** in a state in which the position thereof in the discharge direction is regulated within a region upstream of the

second intersection point **800** in the discharge direction as indicated by a broken line **902** in FIG. **9**, and therefore good alignment can be achieved.

(B) Document of Second Size and Arrangement of Second Projections

[0087] The second projections **103** are provided at positions corresponding to a document of the second size. That is, the second projections **103** are disposed at such positions that a leading end portion of the document of the second size in the discharge direction can be supported by the second inclined surfaces **600**. In other words, the second projections **103** are disposed such that at least part of the second inclined surfaces **600** faces a lower surface of the document as viewed in the Y direction in a state in which a trailing end of the document of the second size discharged onto the discharge tray **303** is abutting the wall surface portion **100** as illustrated in FIG. **10**. In the case where A5R is assumed as the second size, the second vertices **601** of the second projections **103** are preferably disposed such that $L2$ [mm] and $H2$ [mm] satisfy $180 \leq L2 < 240$; and $40 \leq H2 < 60$.

[0088] In the case where the second projections **103** are disposed as described above, the position of a second-size document **1000** disposed onto the discharge tray **303** is regulated by the combination of the swing guide **411**, the first projection **102**, and the second projections **103** as illustrated in FIG. **10**. That is, the second-size document **1000** is discharged while pushing up the swing guide **411** in a state of being nipped by the discharge roller pair **406**, and the leading end portion thereof moves in contact with the second inclined surfaces **600** of the second projections **103**. The second vertices **601** of the second projections **103** are positioned downstream of the first vertex **502** of the first projection **102** in the discharge direction and higher than the first vertex **502** in the gravity direction, and therefore the document **1000** forms an inclined surface shape connecting the first vertex **502** and the second vertices **601**.

[0089] Here, in a state before the first document **1000** is discharged, the space above the discharge tray **303** is blocked by the first projection **102** and the swing guide **411**. Therefore, the first document **1000** is conveyed while pushing up the swing guide **411** and receiving a frictional drag from the swing guide **411** and the first projection **102**. Therefore, even in the case where the conveyance speed of the document **1000** by the discharge roller pair **406** is higher than 340 mm/s, movement of the document **1000** to a position far away from the wall surface portion **100** by an inertia force can be suppressed.

[0090] In the case of discharging the second and subsequent documents **1000**, since the swing guide **411** has already swung to a position higher than the first projection **102**, the influence of the frictional force from the swing guide **411** and the first projection **102** is relatively smaller.

Meanwhile, the document **1000** being discharged does not move to a position far away from the wall surface portion **100** because the document **1000** is discharged while being pushed up along the inclined surface shape formed by the already-discharged document **1000**. As described above, the position of the second-size document **1000** in the discharge direction is regulated by the swing guide **411**, the first projection **102**, and the second projections **103** regardless of in which ordinal number the document **1000** is discharged, and therefore good alignment is achieved.

[0091] To be noted, the angle $\theta 2$ formed between the second straight line $Ln2$ and the horizontal direction is preferably smaller than the angle $\theta 1$ formed between the first straight line $Ln1$ and the horizontal direction, that is, $\theta 2 < \theta 1$ is preferably satisfied. As a result of this, the reaction force that the document **1000** receives when the leading end of the document **1000** comes into contact with the second inclined surfaces **600** projecting to a higher position than the first vertex **502** of the first projection **102** can be attenuated, and thus troubles such as bending or stacking failure of the document **1000** caused as a result of the document **1000** being caught can be suppressed.

[0092] Examples of preferable numerical ranges of the angle $\theta 1$ [deg] formed between the first straight line $Ln1$ and the horizontal direction and the angle $\theta 2$ formed between the second straight line $Ln2$ and the horizontal direction are $40 \leq \theta 1 \leq 55$, $25 \leq \theta 2 \leq 40$, and $\theta 1 - \theta 2 \geq 5$. When $25 \leq \theta 2 \leq 40$ and $\theta 1 - \theta 2 \geq 5$ are satisfied, the document **1000** being caught by the second inclined surfaces **600** can be effectively suppressed. In addition, when $40 \leq \theta 1 \leq 55$ is satisfied, the effect of the first inclined

surface **501** regulating the position of the first-size document **900** as described in (A) above can be enhanced.

(C) Document of Third Size and Arrangement of Third Projections

[0093] The third projections **104** are provided at positions corresponding to a document of a third size. That is, the third projections **104** are disposed at such positions that the third inclined surfaces **700** can support a leading end portion of a document of the third size in the discharge direction. In the case where A4R is assumed as the third size, the third vertices **701** of the third projections **104** are preferably disposed such that $L3$ [mm] and $H3$ [mm] satisfy $240 \leq L3 < 360$ and $60 \leq H3 < 80$.

[0094] In the case where the third projections **104** are disposed as described above, the position of a third-size document **1100** discharged onto the discharge tray **303** is regulated by the cooperation of the swing guide **411**, the first projection **102**, the second projections **103**, and the third projections **104** as illustrated in FIG. **11**. That is, the third-size document **1100** is discharged while pushing up the swing guide **411** in a state of being nipped by the discharge roller pair **406**, and the leading end portion thereof comes into contact with the second inclined surfaces **600** and then moves in contact with the third inclined surfaces **700**. Then, a lower surface of the document **1100** is supported by the first vertex **502** of the first projection **102**, the second vertices **601** of the second projections **103**, and the third vertices **701** of the third projections **104**. Since the third vertices **701** are positioned downstream of the second vertices **601** in the discharge direction and higher than the second vertices **601** in the gravity direction, the document **1100** forms an inclined surface shape smoothly connecting the first vertex **502**, the second vertices **601**, and the third vertices **701**.

[0095] Here, in a state before the first document **1100** is discharged, the space above the discharge tray **303** is blocked by the first projection **102** and the swing guide **411**. Therefore, the first document **1100** is conveyed while pushing up the swing guide **411** and receiving a frictional drag from the swing guide **411** and the first projection **102**. Therefore, even in the case where the conveyance speed of the document **1100** by the discharge roller pair **406** is higher than 340 mm/s, movement of the document **1100** to a position far away from the wall surface portion **100** by an inertia force can be suppressed.

[0096] In the case of discharging the second and subsequent documents **1100**, since the swing guide **411** has already swung to a position higher than the first projection **102**, the influence of the frictional force from the swing guide **411** and the first projection **102** is relatively smaller.

Meanwhile, the document **1100** being discharged does not move to a position far away from the wall surface portion **100** because the document **1100** is discharged while being pushed up along the inclined surface shape formed by the already-discharged document **1100**. As described above, the position of the third-size document **1100** in the discharge direction is regulated by the swing guide **411**, the first projection **102**, the second projections **103**, and the third projections **104** regardless of in which ordinal number the document **1100** is discharged, and therefore good alignment is achieved.

[0097] As described above, the discharge tray **303** of the present exemplary embodiment is configured such that $L1 < L2$, $L4 < L2$, $H1 < H2$, and $H4 < H2$ are satisfied for the swing guide **411**, the first projection **102**, and the second projections **103**. As a result of this, as described in (A) and (B) above, good alignment of first-size and second-size documents in the discharge direction can be maintained. In particular, good alignment of the first-size and second-size documents can be achieved even in the case where the conveyance speed of documents is set to a value higher than 340 mm/s, for example, 400 mm/s or higher, for improving the productivity.

[0098] Further, in the present exemplary embodiment, the third projections **104** are provided on the discharge tray **303** such that $L2 < L3$ and $H2 < H3$ are satisfied. As a result of this, as described in (C) above, good alignments of documents of the third size can be maintained in the discharge direction in addition to the documents of the first size and the second size.

[0099] To be noted, the third projections **104** serving as third supporting portions of the present exemplary embodiment have shapes projecting to higher positions than upper surfaces **107** of the

ribs extending downstream from the second projections **103** in the discharge direction. However, the third projections **104** may be omitted and surfaces **108** which are extensions of the upper surfaces **107** and inclined upward toward the downstream side in the discharge direction may be provided. These surfaces **108** serve as third supporting portions that support the lower surface of a third-size document at a position downstream of the second vertices **601** in the discharge direction and higher than the second vertices **601** in the gravity direction.

Other Embodiments

[0100] In the exemplary embodiment described above, the first projection **102**, the second projections **103**, and the third projections **104** are all formed as rib-like projections extending in the discharge direction. However, each supporting portion may be formed as a surface also extending in the Y direction as long as the supporting portion functions as the first supporting portion, the second supporting portion, or the third supporting portion. For example, a surface having the same outline as the second projections **103** as viewed in the Y direction may be formed on the entirety of the discharge tray **303** in the Y direction, and the first projection **102** may be formed to project from a predetermined position in the Y direction on this surface. In addition, for example, a rib-like projection may be used instead of the upstream supporting surface **101** having a planar shape as the upstream supporting portion to support a sheet.

[0101] In addition, although the discharge tray **303** capable of supporting at least three documents having different lengths in the discharge direction has been described in the exemplary embodiment described above, as a matter of course, the discharge tray may be formed to support documents of a wider variety in size. For example, an extension tray **109** housed in the discharge tray **303** may be pulled out to the downstream side in the discharge direction in FIG. **1** such that a document of a fourth size longer than the third size in the conveyance direction can be supported. For example, the fourth size is an LGL size having a length of 355.6 mm in the conveyance direction. In this case, the downstream end of the extension tray **109** having been pulled out in the discharge direction is preferably higher than the third vertices **701** of the third projections **104**.

[0102] In addition, although a case where the technique of the present disclosure is applied to a sheet discharge configuration of an ADF has been described in the exemplary embodiment described above, the technique of the present disclosure is also applicable to sheet discharging apparatuses other than ADFs, for example, an apparatus that discharges a recording material on which an image has been already formed by an image forming apparatus.

[0103] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0104] This application claims the benefit of Japanese Patent Application No. 2019-102798, filed on May 31, 2019, which is hereby incorporated by reference herein in its entirety.

Claims

1.-14. (canceled)

15. An image reading apparatus comprising: a support tray on which a sheet is supported; a feeding portion configured to feed the sheet supported on the support tray; a reading portion configured to read an image from the sheet fed by the feeding portion; a discharging portion configured to discharge the sheet, from which the image has been read by the reading portion, in a sheet discharge direction; a stack tray on which the sheet discharged by the discharging portion is stacked; and a swing member supported swingably by the support tray and configured to come in contact with an upper surface of the sheet stacked on the stack tray, wherein the stack tray includes a first rib extending in the sheet discharge direction and configured to come in contact with the swing member.

16. The image reading apparatus according to claim 15, wherein the stack tray includes a second rib extending in the sheet discharge direction and disposed at a position separated from the first rib with respect to a sheet width direction (Y) orthogonal to the sheet discharge direction.
